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Does Banking Competition Affect Innovation?

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1 Big picture

- **Question.** Does stronger banking competition increase innovation, or can it actually reduce it once we distinguish between public corporations and private firms?
- **Setting.** The paper studies staggered deregulation of interstate bank branching laws across U.S. states after the Interstate Banking and Branching Efficiency Act (IBBEA). It combines state-level banking deregulation with patent data, patent citations, bank-loan data, and acquisition data.
- **Core challenge.** Innovation and financial development are jointly endogenous. More innovative states may deregulate earlier, and omitted state shocks could affect both bank competition and innovation.
- **Main empirical strategy.** The authors exploit cross-state and over-time variation in the Rice–Strahan branching restrictions index (**RSindex**), control for state and year fixed effects plus a rich set of state controls, and then run event-time and placebo tests to rule out reverse causality and coincident shocks.
- **Main baseline result.** Banking competition *reduces* overall state-level innovation. Moving from the most restrictive state (**RSindex**=4) to the most open state (**RSindex**=0) is associated with about **30.8% fewer patents** and **23.2% fewer patent citations** in the next three years.
- **Ownership decomposition.** The negative average effect is driven by **public corporations**. In the baseline, private firms do not show a direct average response. But once the sample is split by financial dependence, the picture changes sharply.
- **Mechanism 1.** More external-finance-dependent private firms innovate more after deregulation. For these firms, banking competition relaxes financing constraints.
- **Mechanism 2.** Private firms with stronger *out-of-state* bank borrowing before deregulation benefit more after deregulation, consistent with branching reform expanding access to local credit for firms previously underserved by local banks.
- **Mechanism 3.** Public corporations appear to lose access to innovative acquisition targets. After deregulation, small innovative private firms can remain independent instead of selling to public acquirers.
- **Main conceptual message.** The paper is not saying “credit hurts innovation” in a simple aggregate sense. It is saying that bank competition changes *who innovates*. Innovation shifts away from public corporations and toward financially constrained private firms.
- **Why this matters.** A credit-market reform can reallocate innovative activity across organizational forms even if its effect on total patenting is negative.
- **Economic takeaway.** Banking competition is a financing shock, but also a market-for-corporate-control shock. It helps small private innovators finance themselves, which shrinks the supply of innovative targets available to public corporations.

2 Data and measurement (key objects)

2.1 Sample and data sources

- **Time period.** 1976–2006 for the underlying patent data; the main state-year regressions use 1976–2003 because innovation outcomes are measured over the next three years.
- **Main data sources.**
 - NBER Patent Citation Database for patents and citations.
 - Compustat Industrial Annual Files for public-firm controls.
 - DealScan for bank lending relationships.
 - SDC M&A database for acquisition activity.
 - U.S. Census Bureau for the number of private non-farm businesses.
- **Units of observation.**
 - Baseline regressions: **state-year** observations.
 - Mechanism tests: **corporation-year** and **private-firm-year** observations.

2.2 Innovation measurement

- **Patent quantity.** The main quantity measure is the number of patent applications filed in year t that are eventually granted.
- **Patent quality.** The main quality measure is the number of forward citations to those patents.
- **Timing choice.** The paper uses the *application year*, not the grant year, because the application date better captures the timing of the innovative activity.
- **Three-year horizon.** Innovation outcomes are summed over $t + 1$ through $t + 3$ to reflect the long-horizon nature of innovative activity and reduce noise from one-year fluctuations.
- **Truncation corrections.**
 - Patent counts are adjusted using application-grant “weight factors.”
 - Citation counts are adjusted using citation-lag corrections.
- **Final dependent variables.** The baseline outcomes are

$$\text{LnPat} = \log(1 + \text{future patents}), \quad \text{LnCite} = \log(1 + \text{future citations}). \quad (1)$$

- **Additional patent-nature measures.**
 - **Originality:** one minus the Herfindahl index of the technology classes of the patents cited by a patent.
 - **Generality:** one minus the Herfindahl index of the technology classes of patents that cite a patent.

2.3 Public corporations versus private firms

- **Public corporations.** Patent assignees matched to Compustat via the NBER bridge file.
- **Private firms.** Patent assignees that do not match Compustat and are not governments, universities, or individuals.
- **Important distinction.** The paper uses “corporation” for Compustat-listed public firms and “firm” for private firms. That distinction matters throughout the mechanism section.

2.4 Banking competition measure

- **Core variable:** `RSindex`, following Rice and Strahan (2010).
- **Construction.** States can restrict interstate branching using four provisions:
 1. minimum age of target institution,
 2. de novo interstate branching,
 3. acquisition of individual branches,
 4. statewide deposit cap.
- **Scale.** `RSindex` ranges from 0 (most open) to 4 (most restrictive).
- **Interpretation.** A *fall* in `RSindex` means more bank competition. This sign convention is crucial when reading the coefficients: a *positive* coefficient on `RSindex` means deregulation lowers innovation.

2.5 Controls and company-level covariates

- **State-level controls in the baseline.**
 - labor shares in mining, construction, manufacturing, transportation, trade, finance, services, and government,
 - labor-force concentration,
 - nominal gross product,
 - dummies for earlier intrastate and interstate banking deregulation.
- **Corporation-level controls in mechanism regressions.**
 - R&D/assets,
 - ROA,
 - PPE/assets,
 - leverage,
 - capex/assets,
 - industry Herfindahl,
 - Tobin’s Q ,
 - size,
 - age.

2.6 Key descriptive statistics

- **State-year sample.** Average state produces **2,988 patents** and **39,085 citations** over the next three years.
- **Public versus private.** Of the average 2,988 patents, about **1,962** come from public corporations and **1,026** from private firms.
- **Average restriction level.** Mean RSindex is **3.5**.
- **Corporation-year sample.** Average public corporation generates about **5 patents** and **58 citations** over three years.
- **Private-firm-year sample.** Average private firm generates about **0.7 patents** and **7 citations**.

Interpretation. Public corporations account for the larger share of patenting on average, so any negative effect concentrated in public firms can easily dominate a positive effect for a smaller subset of private firms.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline state-level innovation regressions

The core patent-count specification is

$$\text{LnPat}_{i,t+1:t+3} = \alpha + \beta \text{RSindex}_{i,t} + \gamma Z_{i,t} + \text{Year}_t + \text{State}_i + \varepsilon_{i,t}, \quad (1a)$$

and the citation specification is

$$\text{LnCite}_{i,t+1:t+3} = \alpha + \beta \text{RSindex}_{i,t} + \gamma Z_{i,t} + \text{Year}_t + \text{State}_i + \varepsilon_{i,t}. \quad (1b)$$

Interpretation.

- i indexes states and t indexes time.
- State fixed effects absorb time-invariant cross-state differences in innovative intensity.
- Year fixed effects absorb aggregate innovation trends.
- Because a lower RSindex means more competition, a positive β implies that deregulation reduces innovation.
- The horizon $t + 1$ to $t + 3$ turns the regression into a forward-looking reduced-form effect of deregulation on subsequent innovation outcomes.

3.2 Ownership split and patent-nature regressions

The paper re-estimates (1a) and (1b) separately for:

- all firms together,

- public corporations only,
- private firms only.

It also replaces the dependent variable with average patent originality and generality.

Interpretation.

- This is the first decomposition that reveals the negative average effect is driven by public corporations.
- The originality and generality tests ask whether banking competition changes the *type* of innovation rather than just the amount or impact of patented innovation.

3.3 Event-time pretrend specification

To test for reverse causality, the paper estimates

$$\text{LnPat}_{i,t} (\text{LnCite}_{i,t}) = \alpha + \beta_1 \text{Before2+}_{i,t} + \beta_2 \text{Before1}_{i,t} + \beta_3 \text{After1}_{i,t} + \beta_4 \text{After2+}_{i,t} + \text{Year}_t + \text{State}_i + \varepsilon_{i,t}. \quad (2)$$

Interpretation.

- The omitted category is the deregulation year.
- **Before2+** and **Before1** capture pre-deregulation trends.
- **After1** and **After2+** capture post-deregulation effects.
- The key test is whether the before coefficients are near zero while the after coefficients are negative.

3.4 External finance dependence channel

The paper next estimates

$$\text{LnPat}_{i,t+1:t+3} (\text{LnCite}_{i,t+1:t+3}) = \alpha + \beta_1 \text{RSindex}_{k,t} + \beta_2 \text{Dependence}_{i,t} + \beta_3 (\text{RSindex}_{k,t} \times \text{Dependence}_{i,t}) + \gamma Z_{i,k,t} + \text{Ind} \quad (3)$$

Definition of dependence.

- For the main EFD split, **Dependence=1** means *less* external-finance-dependent, and **Dependence=0** means *more* external-finance-dependent.
- For private firms, industry-level EFD is imputed from public firms using patent technology classifications.

Interpretation.

- Because the dependent-variable coding sets the more dependent group at **Dependence=0**, the coefficient β_1 gives the deregulation effect for the more dependent group.
- A negative β_1 means more dependent firms benefit from deregulation because a fall in **RSindex** raises innovation.
- A positive interaction β_3 means the benefit is weaker for less dependent firms.

3.5 Prior banking relationships channel

The prior-lending specification is

$$\text{LnPat}_{i,t+1:t+3} (\text{LnCite}_{i,t+1:t+3}) = \alpha + \beta_1 \text{RSindex}_{k,t} + \beta_2 \text{BankLoans}_{i,t} + \beta_3 (\text{RSindex}_{k,t} \times \text{BankLoans}_{i,t}) + \gamma Z_{i,k,t} + \text{Indus} \quad (4)$$

Interpretation.

- The authors separately compute cumulative *in-state* and *out-of-state* borrowing from DealScan.
- Firms borrowing from out-of-state banks before deregulation are interpreted as firms that were not adequately served by local banking markets.
- If branching deregulation improves local credit access, this group should respond the most.

3.6 Acquirer-side acquisition mechanism

For public corporations, the paper estimates

$$\text{LnPat}_{i,t+1:t+3} = \alpha + \beta_1 \text{RSindex}_{k,t} + \beta_2 \text{Acquisition}_{i,t} + \beta_3 (\text{RSindex}_{k,t} \times \text{Acquisition}_{i,t}) + \gamma Z_{i,k,t} + \text{Industry}_j + \text{Year}_t + \text{Stat} \quad (5)$$

Acquisition intensity is measured three ways:

- dollar volume of acquisitions divided by assets,
- number of deals,
- dollars per deal.

Interpretation.

- If innovation falls because public firms lose innovative targets, the negative deregulation effect should be stronger for firms that rely more on acquisitions.
- A positive interaction β_3 means exactly that: when RSindex falls, innovation drops more for frequent or large acquirers.

3.7 Target-side acquisition mechanism

The target-innovativeness test is

$$\text{TargetInnov}_{i,t} = \alpha + \beta \text{RegChg}_{i,t} + \gamma Z_{i,t} + \text{Firm}_i + \text{Year}_t + \varepsilon_{i,t}, \quad (6)$$

where the dependent variable is either:

- **TargetCount**: total lifetime patents of targets acquired in year t ,
- **TargetCountPerAcq**: the same measure normalized by the number of targets acquired.

The model is estimated with a zero-inflated negative binomial specification.

Interpretation.

- This is a difference-in-differences style test on the innovativeness of targets acquired before versus after deregulation.
- A negative coefficient on **RegChg** means public corporations acquire less innovative targets after deregulation.
- The paper then complements this with an event-time plot of the share of innovative targets in the state, asking whether the supply of innovative targets falls after deregulation.

4 Identification and instruments (what makes the estimates credible)

4.1 Why the deregulation shock is useful

- The identifying variation comes from staggered changes in branching restrictions across states after IBBEA.
- Rice and Strahan (2010) show these regulatory changes increased branch entry and lowered the cost of capital, so the paper treats them as exogenous increases in state-level banking competition.
- Because different states deregulated at different times, the design is better than a single-shock design for dealing with national omitted variables.

4.2 Why the baseline specification is credible

- **State fixed effects** absorb time-invariant state differences such as persistent innovation culture or long-run industrial composition.
- **Year fixed effects** absorb national shocks to innovation and finance.
- **State controls** absorb changes in labor-force composition, concentration, and state economic scale.
- **Earlier deregulation controls** (**Intra**, **Inter**) help isolate the mid-1990s interstate branching reform from earlier banking reforms.

4.3 Reverse-causality tests

- The paper checks whether innovation trends turn before deregulation. They do not.
- In the event-time regression, **Before2+** and **Before1** are insignificant for both patents and citations.
- The paper also restricts the sample to corporations that do not relocate headquarters, or that are headquartered in the state well before deregulation, to reduce concerns that firms move in anticipation of policy changes.

4.4 Omitted-variable tests

- The paper performs a placebo randomization exercise: it preserves the empirical distribution of deregulation years, but randomly reassigns states to those years.
- If the results were really driven by a common omitted shock from the mid-1990s, the randomized treatment should also matter.
- It does not. The placebo **RSindex** coefficients are close to zero and insignificant.

4.5 What makes the mechanism evidence persuasive

- The mechanism tests are not generic kitchen-sink heterogeneity exercises; they follow specific, *ex ante* predictions.
- If the credit-supply interpretation is right, the gains should accrue to firms that are financially constrained or underserved by local banks. That is exactly where the paper finds positive effects.
- If the public-corporation decline comes from a thinner target pool, the effect should be stronger for acquisitive corporations and the innovativeness of acquired targets should fall after deregulation. That is also what the paper finds.

4.6 Main limitations

- **Patents are an incomplete measure of innovation.** The paper can miss nonpatented process innovation or organizational innovation.
- **Private-firm characteristics are partly imputed.** External-finance dependence for private firms is inferred from public-firm industry measures using patent technology mappings.
- **State-level treatment is coarse.** Deregulation is measured at the state level, while innovation and lending relationships may vary more locally within states.
- **Mechanism evidence is indirect.** The paper does not observe a private firm's exact financing contract or reservation value in an acquisition negotiation.
- **Interpretation is U.S.-specific.** The results come from a particular banking reform in a particular institutional setting.

4.7 Relation to the literature

- Relative to the banking-deregulation literature, the paper adds a new real outcome: innovation.
- Relative to the finance-and-innovation literature, it highlights banking competition rather than stock-market development or bankruptcy law.
- Relative to earlier deregulation studies from the 1970s and 1980s, it focuses on the mid-1990s interstate branching changes and emphasizes the reallocation of innovation across public and private organizational forms.

5 Tables and figures

5.1 Table 1 and Figure 1: sample facts and time-series motivation

Table 1

Summary statistics.

This table reports summary statistics for the state-year, public corporation-year, and private firm-year observations in this paper's sample, including dependent, independent, and control variables. Corporation and patent data come from Compustat and the NBER patent database from 1976–2006. We obtain data from DealScan to construct banking relationship measures from 1987–2006 and data from SDC to construct acquisition measures from 1976–2006. Definitions of the variables are in [Appendix A](#).

Panel A: State-year observations						
Variable	p25	Median	Mean	p75	S.D.	N
RSindex	4	4	3.5	4	1.11	1,426
Pat (state)	157	798	2,368	3,166	6,199	1,426
Pat (corporation)	32	305	1,962	1,643	4,483	1,426
Pat (firm)	106	388	1,026	1,209	1,852	1,426
Cite (state)	1,379	7,521	39,085	36,277	95,369	1,426
Cite (corporation)	250	2,531	26,778	18,767	69,552	1,426
Cite (firm)	913	3,848	12,307	13,623	27,510	1,426
Mining (%)	0.16	0.56	3.42	2.56	7.16	1,426
Construction (%)	3.90	4.44	4.62	5.13	1.40	1,426
Manufacturing (%)	12.0	17.2	17.2	23.1	8.0	1,426
Transportation (%)	2.75	3.43	3.64	4.11	1.83	1,426
Trade (%)	13.8	15.3	14.8	16.4	2.6	1,426
Finance (%)	4.28	5.11	5.96	6.38	3.41	1,426
Services (%)	13.4	16.6	17.6	21.1	5.8	1,426
Government (%)	11.2	13.3	14.3	15.8	5.0	1,426
Concentration (%)	11.5	12.5	13.1	13.8	2.9	1,426
Gross product (billion)	25.4	58.0	113.9	136.8	158.4	1,426
Panel B: Public corporation-year observations						
Variable	p25	Median	Mean	p75	S.D.	N
Pat	0	0	4.97	2	15.38	114,937
Cite	0	0	57.76	22.53	178.73	114,937
FFD	-0.63	-0.38	-0.27	-0.11	0.78	114,728
In-state loan (millions)	0	0	7	0	150	79,101
Out-of-state (millions)	0	0	33	0	288	79,101
Assets (millions)	20	99	1,018	625	2,157	84,902
Age	8	17	20.10	31	14.30	88,986
SA index	-15.27	-9.36	-10.58	-5.49	6.08	84,902
R&D/Assets	0	0	0.04	0.03	0.10	133,643
ROA	0.02	0.11	0.02	0.17	0.40	133,643
PP&Assets	0.12	0.26	0.32	0.48	0.25	133,643
Leverage	0.11	0.26	0.31	0.41	0.3	133,643
Capex/Assets	0.02	0.05	0.07	0.09	0.08	133,643
Hindex	0.11	0.18	0.23	0.30	0.18	133,643
TobinQ	0.99	1.3	2.1	2.0	3	133,643
Dollar volume (millions)	0	0	41	0	36.2	119,843
Number of deals	0	0	0.21	0	0.84	119,843
Dollars per deal	0	0	0.02	0	0.41	119,843
Panel C: Private firm-year observations						
Variable	p25	Median	Mean	p75	S.D.	N
Pat	0	0	0.73	1	0.92	463,831
Cite	0	0	7.02	9.25	12.03	463,831
FFD	-0.73	-0.47	-0.31	-0.19	0.91	462,672
In-state loan (millions)	0	0	0.04	0	3.38	342,524
Out-of-state (millions)	0	0	0.36	0	24.01	342,524
TargetCount	0	0	0.33	0	21.1	119,843
TargetCountPerAcq	0	0	0.21	0	11.7	119,843

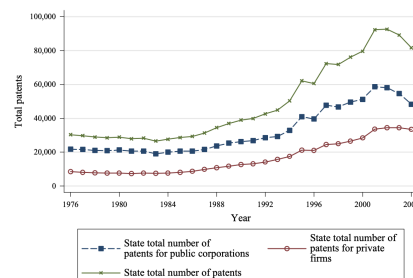


Fig. 1. Patent production through time for corporations and private firms. This figure shows, for each year from 1976–2004, the total number of patents produced by U.S. corporations, the total number of patents produced by private firms in the U.S., and the sum of the two. We identify a patent as being produced by a corporation if it matches to a firm in the Compustat database. We assume a private firm produces the patent if it does not match to a firm in Compustat. We remove from the sample patents produced by universities, governments, and foreign companies.

Read:

- Table 1 establishes the scale of innovative activity and the dominance of public corporations in patenting.
- Figure 1 shows total patenting rises over time, but the public-corporation share appears to decline toward the end of the sample when interstate branching deregulation occurs.
- The private-firm series is smaller in levels but becomes relatively more important near the deregulation period.

Why it matters.

- The figure gives the first hint that aggregate innovation dynamics may hide reallocation across organizational forms.
- Table 1 also tells you why a negative effect on public corporations can dominate the aggregate data.

5.2 Table 2: baseline results and the ownership decomposition

Read:

- For total state innovation, the coefficient on **RSindex** is **0.077** for patents and **0.058** for citations.
- Since deregulation lowers **RSindex** by up to four points, moving from 4 to 0 implies about **30.8% fewer patents** and **23.2% fewer citations** over the next three years.

Table 2

Baseline regressions.

This table reports OLS regression estimates for baseline regressions. Panel A reports the OLS regression estimates of Eqs. (1a) and (1b). The dependent variable in columns 1–3 is the natural logarithm of one plus the sum of the patents generated in a state in the next three years. The dependent variable in columns 4–6 is the natural logarithm of one plus the number of citations for patents generated in a state in the next three years. Panel B reports the OLS regression estimates of Eqs. (1a) and (1b). The dependent variable in columns 1–3 is the patent generality score for patents generated in the next three years. The dependent variable in columns 4–6 is the patent originality score generated in the next three years. Definitions of control variables are in Appendix A. Robust standard errors clustered by year are reported in parentheses. Significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. Observations are at the state-year level from 1976–2003.

	LnPat			LnCite		
	Total (1)	Public (2)	Private (3)	Total (4)	Public (5)	Private (6)
RSindex	0.077*** (0.016)	0.102*** (0.021)	0.030 (0.018)	0.058*** (0.019)	0.066** (0.028)	0.041 (0.024)
Mining	0.013 (0.008)	0.031** (0.015)	0.001 (0.006)	0.013 (0.013)	0.098*** (0.028)	0.000 (0.010)
Construction	−0.012 (0.008)	0.055* (0.029)	−0.028** (0.010)	−0.004 (0.010)	0.115** (0.050)	−0.023 (0.014)
Manufacturing	0.016 (0.010)	0.023 (0.020)	−0.001 (0.007)	0.014 (0.014)	0.077*** (0.023)	−0.000 (0.011)
Transportation	−0.063** (0.029)	−0.139* (0.077)	−0.034* (0.019)	−0.082** (0.031)	−0.190 (0.113)	−0.042* (0.022)
Trade	0.122*** (0.015)	0.116*** (0.035)	0.090*** (0.014)	0.125*** (0.024)	0.172*** (0.058)	0.077*** (0.022)
Finance	−0.005 (0.009)	−0.027 (0.016)	0.048*** (0.007)	−0.017 (0.011)	−0.010 (0.022)	0.037*** (0.009)
Service	−0.090*** (0.010)	−0.103*** (0.022)	−0.083*** (0.008)	−0.104*** (0.016)	−0.060** (0.025)	−0.106*** (0.014)
Government	−0.045*** (0.010)	−0.069*** (0.022)	−0.058*** (0.010)	−0.045*** (0.017)	−0.039 (0.037)	−0.058*** (0.018)
Concentration	0.002 (0.017)	0.075*** (0.020)	0.009 (0.015)	0.069 (0.020)	−0.043** (0.026)	−0.043** (0.019)
Gross product	−0.000 (0.000)	−0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	−0.000 (0.000)	0.000 (0.000)
Intra	0.2226*** (0.040)	−0.402*** (0.060)	−0.058*** (0.020)	−0.251*** (0.038)	−0.468*** (0.067)	−0.098** (0.039)
Inter	−0.095** (0.035)	−0.084 (0.055)	−0.045 (0.038)	−0.166*** (0.044)	−0.216** (0.087)	−0.091* (0.046)
Constant	2.548*** (0.455)	−0.003 (1.232)	3.512*** (0.471)	4.764*** (0.683)	−1.876 (1.572)	6.068*** (0.682)
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,426	1,426	1,426	1,426	1,426	1,426
Adjusted R ²	0.966	0.927	0.971	0.954	0.901	0.954

	Generality			Originality		
	Total (1)	Public (2)	Private (3)	Total (4)	Public (5)	Private (6)
RSindex	−0.002 (0.002)	−0.005 (0.005)	−0.002 (0.003)	−0.002 (0.003)	−0.003 (0.005)	−0.002 (0.003)
Mining	−0.001 (0.002)	0.001 (0.003)	−0.001 (0.002)	−0.000 (0.002)	−0.001 (0.005)	−0.000 (0.002)
Construction	−0.001 (0.002)	−0.000 (0.006)	−0.002 (0.002)	−0.001 (0.004)	−0.003 (0.005)	−0.002 (0.004)
Manufacturing	−0.000 (0.001)	0.002 (0.003)	−0.001 (0.002)	−0.001 (0.002)	−0.004 (0.004)	−0.002 (0.002)
Transportation	−0.001 (0.003)	−0.004 (0.009)	0.004 (0.005)	−0.000 (0.007)	0.005 (0.012)	0.001 (0.006)
Trade	−0.000 (0.003)	0.006 (0.007)	0.001 (0.004)	−0.001 (0.004)	−0.007 (0.008)	−0.000 (0.004)
Finance	−0.004*** (0.001)	0.000 (0.003)	−0.006*** (0.002)	−0.003 (0.002)	−0.005 (0.004)	−0.004** (0.002)
Service	−0.001 (0.003)	−0.009 (0.006)	−0.002 (0.003)	−0.000 (0.002)	0.001 (0.006)	−0.002 (0.002)

- For public corporations, the coefficients are even stronger: **0.102** for patents and **0.066** for citations.
- For private firms, the coefficients are small and statistically insignificant in the baseline.
- Panel B shows no significant effect on patent *generality* or *originality*.

Why it matters.

- This is the central empirical fact of the paper.
- It already reveals that the paper is not about “better ideas versus worse ideas” in a technological-composition sense; the main effect is on the amount and citation impact of patented innovation.
- It also shows the importance of splitting the data by ownership form.

5.3 Tables 3 and 4: no pretrend and no placebo effect

Read:

- Table 3 shows that **Before2+** and **Before1** are insignificant for both patents and citations.
- The negative effects appear only in **After1** and especially **After2+**, and mainly for public corporations.
- Table 4 re-runs the baseline using randomized deregulation years. The placebo **RSindex** coefficients are insignificant across patents and citations.

Table 3
Endogeneity tests.
This table reports OLS regression estimates of Eq. (2). We truncate the state-year panel to a $\pm$ ten-year window around deregulatory events. Definitions of variables are in Appendix A. Robust standard errors clustered by year are reported in parentheses. Significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

	LnPat			LnCite		
	Total (1)	Public (2)	Private (3)	Total (4)	Public (5)	Private (6)
<i>Before</i> ²⁺	0.033 (0.058)	0.060 (0.037)	-0.006 (0.052)	0.014 (0.035)	-0.026 (0.073)	0.066 (0.047)
<i>Before</i> ¹	-0.103 (0.095)	-0.031 (0.032)	-0.111 (0.089)	-0.067 (0.057)	-0.094 (0.131)	-0.019 (0.073)
<i>After</i> ¹	-0.086* (0.050)	-0.081** (0.030)	-0.063 (0.049)	-0.028 (0.049)	-0.056 (0.124)	0.001 (0.036)
<i>After</i> ²⁺	-0.130* (0.070)	-0.051*** (0.029)	-0.100 (0.065)	-0.075** (0.031)	-0.169* (0.083)	-0.045 (0.040)
Constant	0.627*** (0.117)	0.573*** (0.075)	0.012 (0.105)	2.351*** (0.070)	2.432*** (0.146)	-0.132 (0.095)
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	910	910	910	910	910	910
Adjusted R ²	0.936	0.925	0.924	0.940	0.893	0.937

Table 4

Randomization of deregulations.

This table reports OLS regression estimates of Eqs. (1a) and (1b) with randomized state deregulations. The dependent variable in columns 1-3 is the natural logarithm of one plus the sum of the patents generated in the next three years. The dependent variables in columns 4-6 is the natural logarithm of one plus the number of citations for patents generated in the next three years. Definitions of control variables are in Appendix A. Robust standard errors clustered by year are reported in parentheses. Significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. Observations are at the state-year level from 1976-2003.

	LnPat			LnCite		
	Total (1)	Public (2)	Private (3)	Total (4)	Public (5)	Private (6)
<i>RSindex</i>	0.002 (0.038)	-0.028 (0.070)	0.020 (0.027)	-0.007 (0.041)	-0.091 (0.095)	0.019 (0.033)
<i>Mining</i>	0.011 (0.032)	0.030 (0.048)	0.001 (0.027)	0.012 (0.033)	0.068* (0.053)	-0.000 (0.028)
<i>Construction</i>	-0.009 (0.037)	0.059 (0.068)	-0.027 (0.023)	-0.002 (0.043)	0.118 (0.080)	-0.022 (0.033)
<i>Manufacturing</i>	0.017 (0.026)	0.025 (0.044)	0.000 (0.023)	0.015 (0.028)	0.076 (0.054)	0.001 (0.023)
<i>Transportation</i>	-0.069 (0.068)	-0.144 (0.151)	-0.037 (0.038)	-0.086 (0.072)	-0.190 (0.214)	-0.046 (0.041)
<i>Trade</i>	0.130*** (0.061)	0.126 (0.109)	0.093*** (0.041)	0.131* (0.072)	0.177 (0.137)	0.082 (0.054)
<i>Finance</i>	-0.004 (0.028)	-0.025 (0.046)	0.047* (0.027)	-0.016 (0.028)	-0.004 (0.058)	0.037 (0.026)
<i>Service</i>	-0.301*** (0.081)	-0.117* (0.067)	-0.088** (0.038)	-0.112** (0.045)	-0.067 (0.084)	-0.112*** (0.041)
<i>Government</i>	-0.043 (0.036)	-0.066 (0.071)	-0.057*** (0.027)	-0.043 (0.049)	-0.038 (0.095)	-0.057 (0.041)
<i>Concentration</i>	0.009 (0.042)	0.084 (0.067)	-0.026 (0.033)	0.014 (0.046)	0.066 (0.088)	-0.040 (0.037)
<i>Gross product</i>	-0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
<i>Intra</i>	-0.200 (0.123)	-0.373 (0.246)	-0.044 (0.061)	-0.235 (0.163)	-0.465 (0.325)	-0.081 (0.100)
<i>Inter</i>	-0.068 (0.043)	-0.052 (0.096)	-0.032 (0.068)	-0.147*** (0.047)	-0.207 (0.134)	-0.074 (0.096)
Constant	10.433*** (2.208)	8.331* (4.939)	10.781*** (1.426)	9.013*** (2.674)	4.421 (6.466)	10.252*** (1.755)
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,426	1,426	1,426	1,426	1,426	1,426
Adjusted R ²	0.966	0.926	0.971	0.954	0.901	0.954

Why it matters.

- Table 3 addresses reverse causality: innovation does not fall *before* deregulation.
- Table 4 addresses omitted variables: if the mid-1990s simply contained some other shock, the fake deregulation assignments would pick it up. They do not.

5.4 Tables 5 and 6: external-finance dependence channel

Table 5

External financial dependence.

This table reports OLS regression estimates of Eq. (3). We consider companies with EFD values above the median EFD (*Dependence*=0) in year t to be financially dependent. Definitions of control variables are in Appendix A. Robust standard errors clustered by year are reported in parentheses. Significance at the 1%, 5%, and 10% level is indicated by ***, **, and * respectively. Observations are at the firm-year level from 1976-2003.

	Private firms		Public corporations	
	LnPat (1)	LnCite (2)	LnPat (3)	LnCite (4)
<i>RSindex</i>	-0.019*** (0.003)	-0.016** (0.007)	-0.021 (0.029)	-0.060 (0.047)
<i>Dependence</i>	-0.228*** (0.013)	-0.315*** (0.028)	-0.460* (0.232)	-0.605* (0.307)
<i>RSindex</i> $\times$ <i>Dependence</i>	0.046*** (0.004)	0.049*** (0.007)	0.136** (0.062)	0.194** (0.094)
Constant	1.250*** (0.134)	2.349*** (0.361)	0.236** (0.112)	0.347** (0.166)
Controls	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	223,655	223,655	76,015	76,015
Adjusted R ²	0.199	0.085	0.126	0.154

Table 6

External financial dependence with alternative proxies: Assets, age, bank-dependencies and SA index.

This table reports OLS regression estimates of Eq. (3). Panel A defines companies with asset values below the median asset value in year t as financially dependent (*Dependence*=0). Panel B defines companies with age below the median age value in year t as financially dependent (*Dependence*=0). Panel C defines companies with accumulative bank loans (both in-state and out-of-state) above the median accumulative bank loans in year t as financially dependent (*Dependence*=0). Panel D defines companies with SA index below the median SA index in year t as financially dependent (*Dependence*=0), where $SA\ index = -0.737 \times \ln(\text{assets}) + 0.041 \times \ln(\text{assets})^2 - 0.04 \times \text{age}$. Definitions of control variables are in Appendix A. Robust standard errors clustered by year are reported in parentheses. Significance at the 1%, 5%, and 10% level is indicated by ***, **, and * respectively. Observations are at the firm-year level from 1976-2003.

	Panel A: Assets		Panel B: Age		Panel C: Bank dependence		Panel D: SA index	
	LnPat (1)	LnCite (2)	LnPat (3)	LnCite (4)	LnPat (5)	LnCite (6)	LnPat (7)	LnCite (8)
<i>RSindex</i>	-0.009*** (0.003)	-0.017*** (0.007)	-0.012** (0.006)	0.004 (0.012)	-0.018*** (0.003)	-0.033*** (0.008)	-0.012*** (0.004)	0.004 (0.007)
<i>Dependence</i>	-0.288*** (0.013)	-0.517*** (0.027)	-0.053*** (0.026)	0.043 (0.065)	-0.560*** (0.030)	-1.043*** (0.060)	-0.064*** (0.014)	0.019 (0.026)
<i>RSindex</i> $\times$ <i>Dependence</i>	0.039*** (0.003)	0.055*** (0.007)	0.029*** (0.007)	0.010 (0.017)	0.078*** (0.007)	0.137*** (0.035)	0.029*** (0.004)	0.012* (0.007)
Constant	1.263*** (0.143)	2.795*** (0.364)	1.198*** (0.230)	2.589*** (0.733)	1.464*** (0.140)	3.191*** (0.399)	1.201*** (0.135)	2.609*** (0.358)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	223,655	223,655	223,655	223,655	223,655	223,655	223,655	223,655
Adjusted R ²	0.205	0.089	0.198	0.084	0.211	0.094	0.198	0.072

Read:

- In Table 5, for private firms the **RSindex** coefficient is -0.019 for patents and -0.016 for citations, with positive interactions for the less-dependent group.
- That means the gain from deregulation is concentrated among the *more* external-finance-dependent private firms.
- Quantitatively, a fully open state implies about **7.6% more patents** for financially dependent private firms.
- For public corporations, the direct effect for the more dependent group is statistically weak; the interaction suggests that the average negative public-corporation effect is concentrated in the less-dependent group.

- Table 6 shows the same qualitative result when the paper proxies dependence using firm size, age, bank dependence, and the Hadlock-Pierce SA index.

Why it matters.

- These tables are the cleanest evidence that the reform really worked through credit access for a subset of private firms.
- They also explain why the average effect on private firms is near zero: some private firms benefit a lot, but not all private firms are credit constrained.

5.5 Table 7: prior banking relationships

Read:

- For private firms with *in-state* prior loans, the direct **RSindex** effect is not significant.
- For private firms with *out-of-state* prior loans, the **RSindex** coefficient is -0.021 for patents and -0.040 for citations, both highly significant.
- Public corporations do not display the same pattern.

Why it matters.

- This table sharpens the credit-supply mechanism. The firms that gain are not just “financially dependent” in a broad industry-level sense; they are specifically the firms that seem unable to satisfy their credit demand locally before deregulation.
- The absence of a comparable response for public corporations is consistent with the idea that local banking frictions matter much more for private firms.

5.6 Tables 8 and 9, and Figure 2: acquisition mechanism

Read:

- Table 8 shows that the interaction between acquisition intensity and **RSindex** is positive for **dollar volume** (0.040) and **dollars per deal** (0.044), meaning the negative innovation effect of deregulation is stronger for acquisitive corporations.
- Table 9 shows that the targets acquired after deregulation are less innovative: **RegChg** is -0.070 for total target patents and -0.056 for patents per acquisition.
- The paper interprets the first number as acquired targets producing about **21% fewer lifetime patents** after deregulation.
- Figure 2 shows a monotonic decline in the innovative target ratio around the deregulation event.

Why it matters.

- These are the key pieces connecting the positive private-firm story to the negative public-corporation story.
- The paper’s broader message is that banking competition can strengthen stand-alone private innovation while weakening innovation produced through the acquisition channel inside public

Table 7

Banking relationships.

This table reports OLS regression estimates of Eq. (4). Panel A shows the results for private firms and Panel B shows the results for public corporations. *BankLoans* is an indicator taking a value of one if the cumulative dollar value of loans borrowed by company i up to year t is greater than the median in that year. We set cumulative loans to zero for companies that do not have any bank loans prior to the passage of any legislation enforcing IBBEA changes in the companies' headquarters states. In columns 1–2 we consider only the loans borrowed by company i by banks located in the same state as company i 's headquarters. In columns 3–4 we consider only the loans borrowed by company i from banks outside company i 's headquarters state. Definitions of control variables are in Appendix A. Robust standard errors clustered by year are reported in parentheses. Significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. Observations are at the firm-year level from 1976–2003.

Panel A: Private firms				
	In-state		Out-of-state	
	<i>LnPat</i> (1)	<i>LnCite</i> (2)	<i>LnPat</i> (3)	<i>LnCite</i> (4)
<i>RSIndex</i>	–0.003 (0.002)	0.002 (0.007)	–0.021*** (0.003)	–0.040*** (0.008)
<i>BankLoans</i>	–0.381*** (0.020)	–0.652*** (0.056)	–0.561*** (0.031)	–1.050*** (0.061)
<i>RSIndex × BankLoans</i>	0.016*** (0.011)	0.038 (0.025)	0.079*** (0.016)	0.148*** (0.033)
Constant	1.377*** (0.145)	3.038*** (0.386)	1.465*** (0.147)	3.179*** (0.394)
Controls	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	223,655	223,655	223,655	223,655
Adjusted R^2	0.207	0.092	0.211	0.094
Panel B: Public corporations				
	In-state		Out-of-state	
	<i>LnPat</i> (1)	<i>LnCite</i> (2)	<i>LnPat</i> (3)	<i>LnCite</i> (4)
<i>RSIndex</i>	0.020 (0.017)	0.009 (0.032)	0.008 (0.015)	–0.020 (0.021)
<i>BankLoans</i>	–0.130* (0.073)	–0.184* (0.104)	–0.745*** (0.040)	–1.077*** (0.087)
<i>RSIndex × BankLoans</i>	0.028 (0.021)	0.036 (0.032)	0.050*** (0.011)	0.085*** (0.022)
Constant	1.019*** (0.167)	2.428*** (0.282)	1.436*** (0.162)	3.022*** (0.269)
Controls	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	44,702	44,702	44,702	44,702
Adjusted R^2	0.130	0.162	0.148	0.173

Table 9

Patents from acquisition targets.

This table reports zero-inflated negative binomial marginal effect estimates from Eq. (6). The dependent variable in column 1 is *TargetCount*, which we calculate in two steps. First, we calculate the sum of all patents generated by a target during the 1976–2006 sample. Then we sum the number of patents acquired by all the targets of corporation i in the year that the targets are acquired. In column 2 the dependent variable is *TargetCountPerAcq*, which is *TargetCount* divided by the number of targets corporation i acquires in year t . *RegChg* is a dummy variable set to one in the year of (and all years following) a regulation change. We repeat the observations of states with multiple regulation changes for each regulation change. We de-mean the independent variables and include year fixed effects. Significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

	<i>TargetCount</i> (1)	<i>TargetCountPerAcq</i> (2)
<i>RegChg</i>	–0.070* (0.036)	–0.056*** (0.028)
Controls	Yes	Yes
Year FE	Yes	Yes
Firm FE	Yes	Yes
Observations	151,883	151,883

Table 8

Dollar volume of acquisitions.

This table reports OLS regression estimates from Eq. (5) using the natural logarithm of one plus the sum of the patents generated in the next three years as the dependent variable. The acquisition measure, *Acquisition*, for column 1 is the dollar volume spent on acquisitions for corporation i in year t , divided by the corporation's total assets in year t . In column 2 *Acquisition* is the number of acquisitions made by corporation i in year t . In column 3 *Acquisition* is the dollar volume spent on acquisitions for corporation i in year t divided by total assets of corporation i in year t and the number of acquisitions made by corporation i in year t . We de-mean both *RSIndex* and the *Acquisition* measures for ease of interpretation. Definitions of control variables are in Appendix A. Robust standard errors clustered by year are reported in parentheses. Significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. The sample includes Compustat firms from 1976–2003 and the observations are at the firm-year level.

	<i>Dollar volume</i> (1)	<i>Number of deals</i> (2)	<i>Dollar per deal</i> (3)
<i>RSIndex</i>	0.008* (0.004)	0.007 (0.004)	0.008* (0.004)
<i>Acquisition</i>	0.008 (0.014)	0.115*** (0.020)	–0.000 (0.016)
<i>Acquisition × RSIndex</i>	0.040** (0.018)	0.015 (0.010)	0.044** (0.021)
Constant	–2.259*** (0.071)	–2.263*** (0.072)	–2.258*** (0.071)
Controls	Yes	Yes	Yes
State FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
Observations	72,093	72,093	72,093
Adjusted R^2	0.420	0.421	0.420

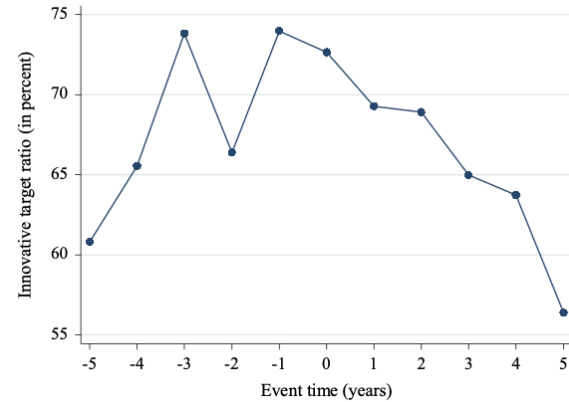


Fig. 2. Innovative target ratio of acquisition targets across state regulation changes. This figure plots states' average Innovative target ratios (*ITR*) from five years before to five years after a state-level legislative change. We calculate *ITR_t* by dividing the number of targets headquartered within a state with at least one patent filing in year t by the total number of private firms in that state-year. We transform the state-year observations to event time, with a state legislative change following IBBEA being an event. States with multiple legislative changes appear in the sample multiple times, once for each event.

corporations.

6 Short summary

- **Method.** Use staggered interstate branching deregulation across states as a quasi-natural experiment for banking competition.
- **Main reduced-form result.** More bank competition lowers aggregate state-level patenting and citations.
- **Who drives it?** Public corporations drive the decline.
- **Who benefits?** Financially constrained private firms and firms previously borrowing from out-of-state banks innovate more after deregulation.
- **Main mechanism.** Better credit access lets innovative private firms remain independent instead of being acquired, reducing the supply of innovative targets for public corporations.
- **Bottom line.** Banking competition reallocates innovation from public corporations toward selected private firms, and the negative public-corporation effect dominates in the aggregate state data.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that interstate branching deregulation in the mid-1990s did *not* raise overall state-level innovation. Instead, aggregate patenting and patent citations fell, mainly because public corporations headquartered in deregulating states became less innovative. At the same time, selected private firms — especially those that were financially constrained or underserved by local banks — became more innovative.

7.2 Economic intuition

The economic logic is two-sided. More banking competition expands credit supply and lowers borrowing costs. That helps small private firms with innovative ideas finance themselves. But this very success changes the acquisition market. If small innovative firms can fund themselves independently, they are less likely to sell to public corporations. Since a meaningful share of public-corporation innovation comes through acquisitions, public-corporation patenting falls. The aggregate state-level effect ends up negative because the public-corporation decline is large.

7.3 Takeaway

A useful way to remember the paper is:

Banking competition does not simply create more or less innovation. It changes the *organizational location* of innovation.

That is why the paper finds both a positive private-firm credit channel and a negative public-corporation acquisition channel at the same time.